Y3S2 Group Project Explaining the Code

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1 Pre-processing

1.1 Atmospheric Assumptions and Composition

When forming a model absorption spectrum, our script takes the input planetary parameters and assumes a one-dimensional, isothermal atmosphere with a constant composition ratio, where pressure (P) decreases with radial distance (r) according to $P(r) \propto \frac{1}{r^2}$. The atmosphere is assumed to behave as an ideal gas, so the number density (n) is calculated as $n = \frac{P}{k_B T}$ where k_B is the Boltzmann constant and T is the temperature. The cloud deck (at a specified pressure) is treated as a hard cutoff and is assumed to be 100% opaque below that level. The opacity is calculated wavelength-by-wavelength from the cross-section files for each gas. These values are interpolated and then weighted by number density and mixing ratio before being combined to calculate the transit depth at each wavelength.

1.2 Radiative Transfer and Optical Depth

When the planet passes in front of the star, light passes through different layers of the atmosphere or hits the opaque interior (clouds or solid surface). Along each light ray, the radiative transfer equation is seen in equation 1

$$I_\lambda(s) = I_{\lambda,0} e^{-\tau_\lambda(s)} \tag{1}$$

where $I_{\lambda,0}$ is the incoming specific intensity from the star, s is the distance along the ray, and τ_λ is the optical depth.

We therefore use the planetary parameters and cross sections to change the wavelength-dependent optical depth, $d\tau_\lambda = \kappa_\lambda \, \rho \, ds = \sum_i n_i \, \sigma_{\lambda,i} \, ds$, where κ_λ is the mass absorption coefficient, ρ is the density, n_i is the number density of species i and $\sigma_{\lambda,i}$ is its absorption cross section.

1.3 Pressure Grid and Hydrostatic Equilibrium

The code represents the atmosphere on a 1D pressure grid from $P = 10^{-4}\,\text{Pa}$ to $P = 10^{7}\,\text{Pa}$. For each pressure level P_j it assigns a temperature $T_j = T_p P_j$

and number density $n_j = \frac{P_j}{k_B T_j}$

To know how far rays travel through each layer, we need to find the radius r at each pressure level. This is obtained by integrating the equation of hydrostatic equilibrium, $\frac{dP}{dr}$, which leads to the expression in equation 2.

$$dr = -\frac{k_B T}{\mu g} dlnP \tag{2}$$

The calculation of gravity relative to radius can stated as $g(r) = g_{\rm ref} \left(\frac{r_{\rm ref}}{r}\right)^2$, where $g_{\rm ref}$ is the surface gravity at a reference pressure $P_{\rm ref}$ and $r_{\rm ref} = R_{\rm p}$ is the corresponding radius. From this reference point, the script steps outwards and inwards in even steps of $\ln P$ to build the arrays $r(P_j)$ and $g(P_j)$. This converts the original pressure grid into a grid of atmospheric shells.

We then set the composition of the atmosphere in terms of logarithmic mixing ratios ($\log_{10} X_i$) for each molecule. This was calculated as a typical subneptune's composition [INSERT REFERENCE]. From these mixing ratios, the mean molecular weight is calculated as $\mu = \sum_i X_i \, m_i$.

The absorption cross sections ($\sigma_{\lambda,i}(P,T)$) are loaded as arrays and we use RegularGridInterpolator to evaluate the cross sections at the pressures and temperatures of the model atmosphere and at the wavelengths of interest. For each wavelength λ_k and layer j, the total absorption per unit path length is then calulated as $\sum_i n_{i,j} \, \sigma_{\lambda_k,i}(P_j, T_j) = n_j \sum_i X_i \, \sigma_{\lambda_k,i}(P_j, T_j)$.

1.4 Collision-Induced Absorption

We then also account for collision-induced absorption (CIA) from H_2–H_2 and He–H_2 pairs. CIA occurs when two colliding molecules form an interaction with an induced dipole moment, allowing transitions to absorb radiation. This is important in H_2-rich atmospheres because H_2 has no permanent dipole and therefore weak intrinsic rotational-vibrational line absorption on its own. Unlike narrow molecular line features, CIA typically contributes a broad continuum-like opacity that fills spectral windows between molecular bands. [INSERT REFERENCE FOR CIA CROSS SECTIONS] Because CIA is a two-body process, the contribution to τ_λ scales as n_i^2 rather than n_i.

$$\tau_\lambda^{\rm CIA} \propto n_j^2 \, X_{\rm H_2}^2 \, \sigma_\lambda^{\rm H_2-H_2} + n_j^2 \, X_{\rm H_2} X_{\rm He} \, \sigma_\lambda^{\rm He-H_2} \tag{3}$$

After these contributions are combined, the code holds a 2D array opacity[k,j] which represents $\sum_i n_{i,j} \sigma_{\lambda_k,i}$ for each wavelength and pressure level.

1.5 Slant Optical Depth and Transit Integration

Given the radial grid r_j and the opacity at each layer, the next part of our script finds the optical depth $\tau_\lambda(r)$ along rays that pass through the planet at different layers r. For a ray with minimum radius r_i, the path length through layers at larger radii is set by equation 4 so that the path length across a shell between r_j and r_{j+1} is $\Delta s_j = s(r_{j+1}) - s(r_j)$.

$$s(r) = \sqrt{r^2 - r_i^2} \tag{4}$$

The code loops over the layers above each impact parameter and approximates the slant optical depth by the sum 5 where the opacity is averaged between adjacent layers to obtain an estimate.

$$\tau_\lambda(r_i) \approx \sum_j \left[\frac{\text{opacity}_{k,j} + \text{opacity}_{k,j+1}}{2} \right] \Delta s_j \tag{5}$$

If the pressure at a given radius exceeds the prescribed cloud-top pressure P_{cloud}, a very large additional optical depth (e.g. $\tau \sim 10^3$) is added to mimic an effectively opaque cloud deck.

The fraction of light transmitted along that ray is $e^{-\tau_\lambda(r_i)}$. The script converts this to an effective transit radius with the standard annulus-integration formula for transmission spectroscopy in equation 6.

$$\Delta_\lambda = \frac{\pi R_{\text{p}}^2 + 2 \int_{r>R_{\text{p}}} r \left[1 - e^{-\tau_\lambda(r)}\right] \mathrm{d}r - 2 \int_{r<R_{\text{p}}} r\, e^{-\tau_\lambda(r)}\, \mathrm{d}r}{\pi R_{\text{s}}^2}. \tag{6}$$

The first term represents the area of an opaque disk of radius R_{p}. The second term adds on the contribution from atmospheric annuli above this radius, weighted by the fraction of light they block. The third term subtracts light that is able to leak through layers that are geometrically inside R_{p} but not completely opaque because of the finite optical depth. In practice these integrals are again evaluated numerically using the discrete r_i grid and the precomputed values of $e^{-\tau_\lambda(r_i)}$.

The result is a model transit depth Δ_λ as a function of wavelength, which the notebook plots in parts-per-million (ppm). Changing the planetary radius, mass, temperature, mean molecular weight, or cloud-top pressure directly affects the computed scale height and optical depths, and therefore changes the amplitude and shape of the spectral features.